Algebra II with Trigonometry Chapter 7.1

Olympiad Solutions

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Solutions

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1. Without using a calculator, compute $\frac{\sin 20^\circ + \sin 40^\circ + \sin 80^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 80^\circ}$.

Let $S = \frac{\sin 20^\circ + \sin 40^\circ + \sin 80^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 80^\circ}$.

Let's use the sum-to-product identities:

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

So,

$$\sin 20^\circ + \sin 80^\circ = 2\sin 50^\circ \cos 30^\circ$$

Thus,

$$\sin 20^\circ + \sin 40^\circ + \sin 80^\circ = [\sin 20^\circ + \sin 80^\circ] + \sin 40^\circ = 2\sin 50^\circ \cos 30^\circ +$$

But notice that a similar approach can be taken for the denominator:

$$\cos 20^\circ + \cos 80^\circ = 2\cos 50^\circ \cos 30^\circ$$

So,

$$\cos 20^\circ + \cos 40^\circ + \cos 80^\circ = 2\cos 50^\circ \cos 30^\circ + \cos 40^\circ$$

Therefore,

$$S = \frac{2\sin 50^\circ \cos 30^\circ + \sin 40^\circ}{2\cos 50^\circ \cos 30^\circ + \cos 40^\circ}$$

Now, use $\sin 50^\circ = \cos 40^\circ$ and $\cos 50^\circ = \sin 40^\circ$:

So numerator:

$$2\sin 50^\circ \cos 30^\circ + \sin 40^\circ = 2\cos 40^\circ \cos 30^\circ + \sin 40^\circ$$

Denominator:

$$2\sin 40^\circ \cos 30^\circ + \cos 40^\circ$$

Take out common factors where possible.

Let's try to numerically simplify for elegance:

Note:

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

So numerator:

$$2\cos 40^\circ \cdot \frac{\sqrt{3}}{2} + \sin 40^\circ = \sqrt{3}\cos 40^\circ + \sin 40^\circ$$

$$= \sin 40^\circ + \sqrt{3}\cos 40^\circ$$

Denominator:

$$2\sin 40^\circ \cdot \frac{\sqrt{3}}{2} + \cos 40^\circ = \sqrt{3}\sin 40^\circ + \cos 40^\circ$$

Thus,

$$S = \frac{\sin 40^\circ + \sqrt{3}\cos 40^\circ}{\sqrt{3}\sin 40^\circ + \cos 40^\circ}$$

Divide numerator and denominator by $\cos 40^\circ$:

$$\frac{\tan 40^\circ + \sqrt{3}}{\sqrt{3}\tan 40^\circ + 1}$$

Let $t = \tan 40^\circ$:

$$S = \frac{t + \sqrt{3}}{\sqrt{3}t + 1}$$

Now, recognize that:

$$\tan(60^\circ - 40^\circ) = \tan 20^\circ = \frac{\tan 60^\circ - \tan 40^\circ}{1 + \tan 60^\circ \tan 40^\circ}$$

But $\tan 60^\circ = \sqrt{3}$, so:

$$\tan 20^\circ = \frac{\sqrt{3} - t}{1 + \sqrt{3}t}$$

But observe that

$$S = \frac{t + \sqrt{3}}{\sqrt{3}t + 1} = (\text{interchange numerator and denominator in the formula al}$$

Notice that

$$\frac{t + \sqrt{3}}{\sqrt{3}t + 1} = \cot 20^\circ$$

Because:

$$\cot 20^\circ = \frac{1}{\tan 20^\circ}$$

So:

$$\frac{1}{\tan 20^\circ} = \frac{1 + \sqrt{3}t}{\sqrt{3} - t}$$

But upon closer check, this is not precisely equal unless we made a sign error or variable flip. Let's check this:

Cross-multiplied:

$$(t+\sqrt{3})(\sqrt{3}-t) = (\sqrt{3}t+1)(1+\sqrt{3}t)$$

Multiply out:

$$(t+\sqrt{3})(\sqrt{3}-t) = (\sqrt{3}t+1)(1+\sqrt{3}t)$$

LHS:

$$t\sqrt{3} - t^2 + 3 - \sqrt{3}t = 3 - t^2$$

RHS:

$$(\sqrt{3}t+1)(1+\sqrt{3}t) = \sqrt{3}t + (\sqrt{3}t)^2 + 1 + \sqrt{3}t = 1 + 2\sqrt{3}t + 3t^2$$

Wait: maybe another approach is preferable.

Let’s pick a value.

Try $x = 30^\circ$:

Well, actually, let's try to check with $x = 40^\circ$:

$$\tan(60^\circ - 40^\circ) = \tan 20^\circ = \frac{\tan 60^\circ - \tan 40^\circ}{1 + \tan 60^\circ \tan 40^\circ} = \frac{\sqrt{3}-t}{1+\sqrt{3}t}$$

Let’s consider the reciprocal:

$$\cot 20^\circ = \frac{1+\sqrt{3}t}{\sqrt{3}-t}$$

Thus, turning this around,

$$\Rightarrow t = \tan 40^\circ, \quad \frac{t+\sqrt{3}}{\sqrt{3}t+1} =$$

Let's try plugging $t = \tan 40^\circ \approx 0.8391$:

$$\frac{0.8391 + 1.7321}{1.7321 \times 0.8391 + 1} = \frac{2.5712}{2.4538} \approx 1.048$$

What is $\tan 46^\circ$? $\tan 46^\circ \approx 1.0355$

But $\tan 60^\circ = \sqrt{3} \approx 1.732$ $\tan 70^\circ \approx 2.747$ But $\cot 20^\circ \approx 2.747$.

So our expression numerically does not match $\tan 60^\circ$ nor $\tan 20^\circ$ nor $\cot 20^\circ$. Our algebra is fine but the expression simplifies to a number.

Thus, the value is:

$$S = \frac{\sin 20^\circ + \sin 40^\circ + \sin 80^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 80^\circ} = \frac{\tan 40^\circ + \sqrt{3}}{\sqrt{3} \tan 40^\circ + 1}$$

Now, let's notice that

$$\frac{a + b}{ba + 1}$$

If we take $\tan(60^\circ + 40^\circ)$:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

But that's not helpful with these numbers.

Alternatively, let's check

Alternatively, compute numerically:

$$\sin 20^\circ = 0.3420 \sin 40^\circ = 0.6428 \sin 80^\circ = 0.9848$$

Sum: $0.3420 + 0.6428 + 0.9848 = 1.9696$

$$\cos 20^\circ = 0.9397 \cos 40^\circ = 0.7660 \cos 80^\circ = 0.1736$$

Sum: $0.9397 + 0.7660 + 0.1736 = 1.8793$

Their quotient:

$$\frac{1.9696}{1.8793} \approx 1.048$$

But note that $\tan 47^\circ = 1.072$, close, suggesting $S \approx \tan 47^\circ$.

Let's try a clever identity.

Notice that:

Try to write numerator and denominator as $A \sin x + B \cos x$:

Alternatively, note that

$\tan x + \tan(60^\circ - x) + \tan(60^\circ + x) = \tan x + \tan(60^\circ - x) + \tan(60^\circ + x)$

This is a known identity (see here):

$$\tan\theta + \tan(60^\circ - \theta) + \tan(60^\circ + \theta) = 3\tan 3\theta$$

But our numbers: Let $x = 20^\circ$, then

$$\tan 20^\circ + \tan 40^\circ + \tan 80^\circ = 3\tan 60^\circ = 3\sqrt{3}$$

But that's for tangents, not sines and cosines.

Alternatively, express numerator and denominator as imaginary parts of complex exponentials. Set $z = e^{i20^\circ}$, then $\sin 20^\circ = \Im(z)$, $\sin 40^\circ = \Im(z^2)$, $\sin 80^\circ = \Im(z^4)$

Sum: $\Im(z + z^2 + z^4)$

Similarly, $\cos 20^\circ + \cos 40^\circ + \cos 80^\circ = \Re(z + z^2 + z^4)$

Therefore,

$$S = \frac{\Im(z + z^2 + z^4)}{\Re(z + z^2 + z^4)}$$

But

$$z + z^2 + z^4 = e^{i20^\circ} + e^{i40^\circ} + e^{i80^\circ}$$

Sum geometric series:

First, let's recognize that $e^{i20^\circ} + e^{i40^\circ} + e^{i80^\circ}$ Actually, let's try another trigonometric identity.

Alternatively, accept the previous algebraic simplification:

$$S = \frac{\tan 40^\circ + \sqrt{3}}{\sqrt{3}\tan 40^\circ + 1}$$

Since $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$, but that's not helpful unless denominator is subtraction.

Alternatively, pose the answer as

$$\boxed{\frac{\tan 40^\circ + \sqrt{3}}{\sqrt{3}\tan 40^\circ + 1}}$$

Alternatively, numerically, this is ≈ 1.048. Compare with $\tan 47^\circ \approx 1.072$, so it's just below that.

Thus, the simplest exact form is:

$$\boxed{\frac{\tan 40^\circ + \sqrt{3}}{\sqrt{3}\tan 40^\circ + 1}}$$

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2. Simplify $\frac{1-\tan^2 x}{1+\tan^2 x}$ without converting to sines and cosines, and find its value when $x = 22.5^\circ$.

Let $y = \frac{1-\tan^2 x}{1+\tan^2 x}$.

Recall that

$$\tan 2x = \frac{2\tan x}{1 - \tan^2 x} \implies 1 - \tan^2 x = \frac{2\tan x}{\tan 2x}$$

But we are not to use sines and cosines.

Alternatively, recall that for any x,

$$\cos 2x = 1 - 2\sin^2 x = 2\cos^2 x - 1 = \frac{1-\tan^2 x}{1+\tan^2 x}$$

Thus,

$$\frac{1-\tan^2 x}{1+\tan^2 x} = \cos 2x$$

Now evaluate for $x = 22.5^\circ$:

$$\cos 2x = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Final answers:

$$\boxed{\frac{1-\tan^2 x}{1+\tan^2 x} = \cos 2x}$$

and for $x = 22.5^\circ$,

$$\boxed{\frac{1}{\sqrt{2}}}$$

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3. If $\tan\theta + \cot\theta = 4$, compute the possible values of $\sin 2\theta$.

Let's write:

$$\tan\theta + \cot\theta = 4 \Rightarrow \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 4 \Rightarrow \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = 4 \Rightarrow \frac{1}{\sin\theta\cos\theta}$$

But

$$\sin 2\theta = 2\sin\theta\cos\theta = 2 \times \frac{1}{4} = \frac{1}{2}$$

Thus,

$$\boxed{\frac{1}{2}}$$

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4. If $\sin\alpha + \sin\beta = \sqrt{2}$ and $\cos\alpha + \cos\beta = \sqrt{2}$, find the exact value of $\sin(\alpha+\beta)$.

Sum-to-product:

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

Dividing both equations:

$$\frac{\sin\alpha+\sin\beta}{\cos\alpha+\cos\beta} = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)}{2\cos\left(\frac{\alpha+\beta}{2}\right)} = \tan\left(\frac{\alpha+\beta}{2}\right)$$

Given that both numerators and denominators equal $\sqrt{2}$, so the ratio is 1, so:

$$\tan\left(\frac{\alpha+\beta}{2}\right) = 1 \implies \frac{\alpha+\beta}{2} = 45^\circ \implies \alpha+\beta = 90^\circ$$

Thus,

$$\sin(\alpha+\beta) = \sin 90^\circ = 1$$

$$\boxed{1}$$

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5. Evaluate $\dfrac{\sec A - 1}{\sec A + 1} + \dfrac{\sec A + 1}{\sec A - 1}$ in simplest form for $0 < A < \frac{\pi}{2}$.

Let’s compute:

Let $x = \sec A$

$$E = \frac{x-1}{x+1} + \frac{x+1}{x-1}$$

Combine over a common denominator:

$$E = \frac{(x-1)^2 + (x+1)^2}{x^2-1} = \frac{x^2 - 2x + 1 + x^2 + 2x + 1}{x^2-1} = \frac{2x^2+2}{x^2-1} = \frac{2(x^2 -}{x^2 -}$$

But $x^2 = \sec^2 A = 1 + \tan^2 A$, so $x^2 - 1 = \tan^2 A$:

$$E = \frac{2(1 + \tan^2 A + 1)}{\tan^2 A} = \frac{2(\tan^2 A + 1)}{\tan^2 A} = 2 + \frac{2}{\tan^2 A} = 2 + 2\cot^2 A$$

Final boxed answer:

$$\boxed{2 + 2\cot^2 A}$$

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6. Let $x = 3\sec\theta$ and $0 < \theta < \frac{\pi}{2}$. What is the value of $\sqrt{x^2-9} - \sqrt{x^2-3x}$ in terms of θ?

Given $x = 3\sec\theta$:

First,

$$x^2 = 9\sec^2\theta$$

Now,

$$\sqrt{x^2-9} = \sqrt{9\sec^2\theta - 9} = \sqrt{9(\sec^2\theta - 1)} = 3\sqrt{\sec^2\theta - 1}$$

But $\sec^2\theta - 1 = \tan^2\theta$, so $3\tan\theta$:

$$\sqrt{x^2-9} = 3\tan\theta$$

Second term:

$$\sqrt{x^2-3x} = \sqrt{9\sec^2\theta - 9\sec\theta} = \sqrt{9(\sec^2\theta - \sec\theta)} = 3\sqrt{\sec^2\theta - \sec\theta}$$

Now,

$$\sec^2\theta - \sec\theta = \sec\theta(\sec\theta - 1)$$

But $\sec\theta - 1 = \frac{1-\cos\theta}{\cos\theta}$, but better to factor:

Let's write everything in terms of sine and cosine:

Alternatively, leave it as is:

$$\sqrt{\sec^2\theta - \sec\theta} = \sqrt{\sec\theta(\sec\theta - 1)}$$

But perhaps we can relate it to $\tan\theta$:

Note that:

$$(\sec\theta - 1)^2 = \sec^2\theta - 2\sec\theta + 1 \implies \sec^2\theta - \sec\theta = (\sec\theta - 1)^2 + \sec\theta -$$

Alternatively, build upon this:

Let's write $\sec^2\theta - \sec\theta = y$, so

Recall that, for all $0 < \theta < \frac{\pi}{2}, \tan\theta > 0, \sec\theta > 1$

But notice:

$$\sqrt{x^2 - 9} - \sqrt{x^2 - 3x} = 3\tan\theta - 3\sqrt{\sec\theta(\sec\theta - 1)}$$

But $\sec\theta(\sec\theta - 1) = \sec^2\theta - \sec\theta = 1 + \tan^2\theta - \sec\theta$

Let's try substitution $t = \tan\theta$, $\sec\theta = \sqrt{1+t^2}$

So,

$$\sqrt{\sec^2\theta - \sec\theta} = \sqrt{1 + t^2 - \sqrt{1+t^2}}$$

Let's try to relate this further. Set $t = \tan\theta$:

Suppose set $\sec\theta = s$, so $s = \sec\theta > 1$.

Thus,

$$L = 3\tan\theta - 3\sqrt{s(s-1)}$$

Let's try to write $\tan\theta = \sqrt{s^2-1}$:

Let's factor:

$$s(s-1) = s^2 - s$$

So,

$$\tan\theta - \sqrt{s^2-s}$$

But $t = \sqrt{s^2-1}$

Therefore,

$$t - \sqrt{s^2-s} = \sqrt{s^2-1} - \sqrt{s^2-s}$$

Try to rationalize the numerator:

$$\sqrt{s^2-1} - \sqrt{s^2-s} = \frac{(s^2-1)-(s^2-s)}{\sqrt{s^2-1}+\sqrt{s^2-s}} = \frac{-1+s}{\sqrt{s^2-1}+\sqrt{s^2-s}} = \frac{}{\sqrt{s^2-}}$$

Therefore,

$$L = 3\left(\frac{s-1}{\sqrt{s^2-1}+\sqrt{s^2-s}}\right)$$

Recall that $\sec\theta = s$, $t = \sqrt{s^2-1}$

But also,

$$\sqrt{s^2-s} = \sqrt{s^2-s} = \sqrt{s(s-1)}$$

But $s > 1$, so both positive.

Alternatively, try a value to guess the form.

Let's try $\theta = 60^\circ$:

$$\sec 60^\circ = 2, \tan 60^\circ = \sqrt{3}$$

Then, $x = 3 \cdot 2 = 6$

$$\sqrt{x^2 - 9} = \sqrt{36 - 9} = \sqrt{27} = 3\sqrt{3}$$

$$\sqrt{x^2 - 3x} = \sqrt{36 - 18} = \sqrt{18} = 3\sqrt{2}$$

So,

$$L = 3\sqrt{3} - 3\sqrt{2} = 3(\sqrt{3} - \sqrt{2})$$

Meanwhile, $\tan 60^\circ = \sqrt{3}$, $\sec 60^\circ = 2$, $s = 2$,
$t = \sqrt{s^2 - 1} = \sqrt{4 - 1} = \sqrt{3}$. Confirmed.

So,

$$L = 3(\sqrt{3} - \sqrt{2})$$

Alternatively, from above formula,

$$s = 2, L = 3\left(\frac{2 - 1}{\sqrt{2^2 - 1} + \sqrt{2^2 - 2}}\right) = 3\left(\frac{1}{\sqrt{3} + \sqrt{2}}\right)$$

But above, $3(\sqrt{3} - \sqrt{2})$

So,

$$3(\sqrt{3} - \sqrt{2}) = 3 \cdot \frac{(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2})}{\sqrt{3} + \sqrt{2}} = 3 \cdot \frac{3 - 2}{\sqrt{3} + \sqrt{2}} = 3 \cdot \frac{1}{\sqrt{3} + \sqrt{2}}$$

So both expressions are equal.

Therefore, the compact answer is:

$$\boxed{3\left(\tan\theta - \sqrt{\sec\theta(\sec\theta - 1)}\right)}$$

Or, equivalently,

$$\boxed{\dfrac{3(\sec\theta - 1)}{\tan\theta + \sqrt{\sec\theta(\sec\theta - 1)}}}$$

Either boxed form is correct and elegant.